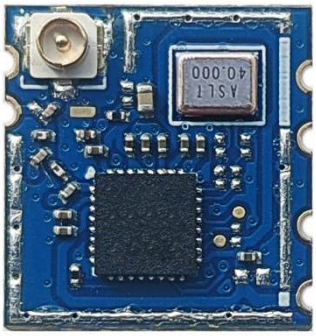
BL-M35343XU1

802.11ax 150Mbps WLAN+BLE+SLE Combo USB Module Specification

SHENZHEN BILIAN ELECTRONIC CO., LTD

Add: 10~11/F, Building 1A, Huaqiang idea park, Guangming district, Shenzhen. Guangdong, China

Web: www.b-link.net.cn



(Top View)



(Bottom View)

Module Name: BL-M35343XU1	
Module Type: 802.11b/g/n/ax 150Mbps WLAN+BLE+SLE Combo USB Module	
Revision: V1.0	
Customer Approval:	
Company:	
Title:	
Signature:	Date:
Approval:	
Title:	
Signature:	Date:

Revision History

Revision	Summary	Release Date	Revised By
1.0	Official release	2024-06-06	Ch

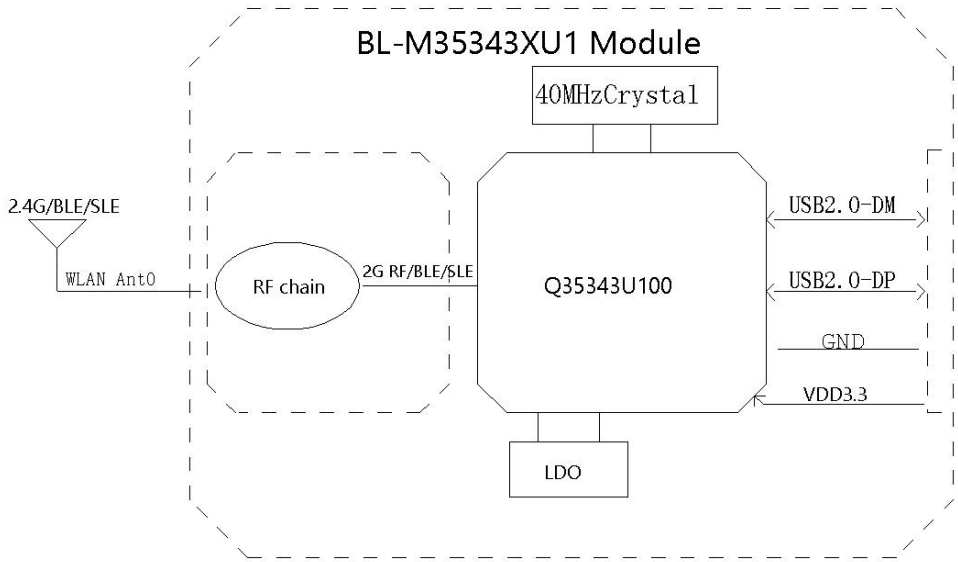
1. Introduction

BL-M35343XU1 is a highly integrated 2.4GHz WLAN+BLE+SLE Combo module base on Q35343U100 single chip, which compatible IEEE802.11b/g/n/ax standard, supports Bluetooth Low Energy 4.2/5.2 and NearLink SLE1.0. This module offering feature-rich wireless connectivity at high standards, delivering reliable, cost-effective and longer communication distance, suitable for AIoT terminals such as consumer IP camera, dashcam, UAV, sweeping robot and OTT boxes.

1.1 Features

- Operating Frequencies: 2.4~2.4835GHz
- Host Interface is USB2.0
- Compatible IEEE802.11 b/g/n/ax, Maximum PHY rate up to 150Mbps
- Support 802.11ax MCS0 up to MCS9 (20MHz band width only)
- Compatible Bluetooth Low Energy 4.0/4.1/4.2/5.0/5.1/5.2
- Compatible NearLink SLE1.0
- Industrial Grade operating temperature range
- Connect to external antenna through IPEX connector
- Single 3.3V±0.2V Power Supply

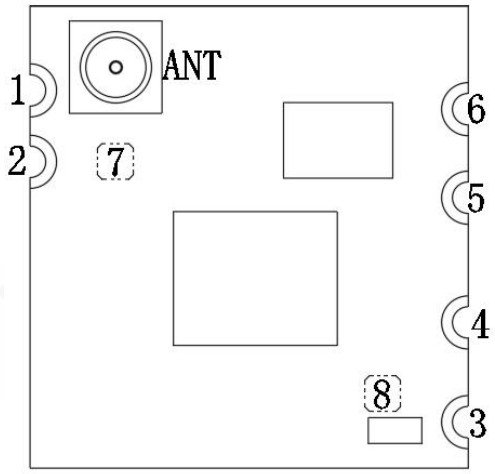
1.2 Block Diagram



1.3 General Specifications

Module Name	BL-M35343XU1
Chipset	Q35343U100
WLAN Standards	IEEE802.11b/g/n/ax
Host Interface	USB2.0
Antenna	Connect to external antenna through IPEX connector
Dimension	13.0*12.3*2.0mm (L*W*H)
Power Supply	3.3V±0.2V @600 mA (Max)
Operation Temperature	-40℃ to +85℃
Operation Humidity	10% to 95% RH (Non-Condensing)

2. Pin Assignments



(Top view)

2.1 Pin Definition

No	Pin Name	Type	I/O Level	Description
1	GND	RF		RF Ground for WLAN_ANT
2	RF	NC		NC (Reserved RF Pad for 2.4G WLAN_ANT/BLE/SLE)
3	VDD3.3	P		DC 3.3V Power Supply
4	USB2.0_DM	A I/O		USB2.0 device Transmitter/Receiver Differential Pair
5	USB2.0_DP	A I/O		USB2.0 device Transmitter/Receiver Differential Pair
6	GND	P		Power Ground

7	GND	P		Auxiliary RF Ground connection PAD for ANT (It can be left floating, but grounding is strongly recommended to enhance ground connectivity and avoid RF interference)
8	GND	/		Auxiliary Ground connection PAD for power and USB (It can be left floating, but grounding is strongly recommended to reduce the power loop area and avoid interference with RF)
	ANT	RF		IPEX for 2.4G WLAN_ANT/BLE/SLE)

P: Power, I: Input, O: Output, I/O: In/Output, RF: Analog RF Port

3. Electrical and Thermal Specifications

3.1 Recommended Operating Conditions

Parameters		Min	Typ	Max	Units
Ambient Operating Temperature		-40	25	85	℃
External Antenna VSWR			1.7	2.0	/
Supply Voltage	VDD3.3	3.1	3.3	3.5	V

3.2 Current Consumption

Conditions : VDD3.3=3.3V, Ta:25℃			
Use Case	VDD3.3 Current		
	Typ(I_{RMS})	Max(I_{Peak})	Units
WLAN/BLE/SLE Unassociated (Android Driver)	59	72	mA
2.4G WLAN TCP throughput TX 74Mbps ((Android, WLAN only)	115	164	mA
2.4G WLAN TCP throughput RX 55Mbps ((Android, WLAN only)	115	216	mA
2.4G 11b@1Mbps TX@20dBm (TX RF test)	365	396	mA
2.4G 11b@1Mbps (RX RF test)	110	130	mA
2.4G 11b@11Mbps TX@20dBm (TX RF test)	358	416	mA
2.4G 11g@6Mbps TX@18dBm (TX RF test)	343	476	mA
2.4G 11g@6Mbps (RX RF test)	110	130	mA
2.4G 11g@54Mbps TX@17dBm (TX RF test)	226	388	mA
2.4G 11n@HT20_MCS0 TX@18dBm (TX RF test)	317	428	mA
2.4G 11n@HT40_MCS0 TX@18dBm (TX RF test)	315	427	mA
2.4G 11n@HT40_MCS7 TX@16dBm (TX RF test)	251	364	mA
2.4G 11ax@HE_SU 20M_MCS0 TX@17dBm (TX RF test)	310	420	mA

2.4G 11ax@HE_SU 20M_MCS9 TX@15dBm (TX RF test)	250	356	mA
BT BLE_1M TX@ 20dBm (RF-Test)	350	436	mA
BT BLE_1M RX (RF-Test)	73	100	mA
BT BLE_2M TX@ 20dBm (RF-Test)	299	460	mA
BT BLE_2M RX (RF-Test)	74	100	mA
SLE 1M GFSK TX@ 20dBm (RF-Test)	210	392	mA
SLE 2M GFSK TX@ 20dBm (RF-Test)	159	392	mA
SLE 2M GFSK RX (RF-Test)	72	100	mA
SLE 1M QPSK：TX@14dBm (RF-Test)	142	296	mA
SLE 2M QPSK：TX@14dBm (RF-Test)	117	304	mA
SLE 2M QPSK RX (RF-Test)	72	100	mA

4. WLAN & Bluetooth & NearLink RF Specifications

4.1 2.4G WLAN RF Specifications

Conditions : VDD3.3=3.3V; Ta:25℃			
Features	**Description**		
WLAN Standard	IEEE 802.11b/g/n/ax		
Frequency Range	2.4~2.4835GHz (2.4GHz ISM Band)		
Channels	Ch1~Ch13 (For 20MHz Channels)		
Modulation	802.11b (DSSS): DBPSK, DQPSK, CCK; 802.11g (OFDM): BPSK, QPSK, QAM16, QAM64; 802.11n (OFDM): BPSK, QPSK, QAM16, QAM64; 802.11ax (OFDMA): BPSK, BPSK_DCM, QPSK, QPSK_DCM, QAM16, QAM16_DCM, QAM64, QAM256		
Data Rate	802.11b: 1, 2, 5.5, 11Mbps; 802.11g: 6, 9, 12, 18, 24, 36, 48, 54Mbps; 802.11n (HT20): MCS0~MCS7_6.5~72.2Mbps; 802.11n (HT40): MCS0~MCS7_13.5~150Mbps; 802.11ax (HE_MU, 26~106RU): MCS0~MCS9_0.4~50Mbps; 802.11ax (HE_SU, non-OFDMA 20MHz): MCS0~MCS9_3.6~114.7Mbps;		
Frequency Tolerance	≦ ±20ppm		
2.4G Transmitter Specifications （TX power tolerance calibrated, customers can define the target TX power within recommended range by modifying configuration file of the driver software ）			
TX Rate	**Recommended Target TX Power（dBm）**	**TX Power Tolerance（dBm）**	**EVM（dB）**

802.11b@1~11Mbps	20	±1.5	≦-10
802.11g@6Mbps	18	±1.5	≦-10
802.11g@54Mbps	17	±1.5	≦-25
802.11n@HT20_MCS0	18	±1.5	≦-10
802.11n@HT20_MCS7	16	±1.5	≦-28
802.11n@HT40_MCS0	18	±1.5	≦-10
802.11n@HT40_MCS7	16	±1.5	≦-28
802.11ax@HE_SU 20M_MCS0	17	±1.5	≦-15
802.11ax@HE_SU 20M_MCS9	15	±1.5	≦-32
2.4G Receiver Specifications			
RX Rate	**Min Input Level (Typ. dBm)**	**Max Input Level (Typ. dBm)**	**PER**
802.11b@1Mbps	-95	-10	< 8%
802.11b@11Mbps	-88	-10	< 8%
802.11g@6Mbps	-92	-10	< 10%
802.11g@54Mbps	-74	-10	< 10%
802.11n@HT20_MCS0	-91	-10	< 10%
802.11n@HT20_MCS7	-72	-10	< 10%
802.11n@HT40_MCS0	-89	-10	< 10%
802.11n@HT40_MCS7	-70	-10	< 10%
802.11ax@HE_SU 20M_MCS0	-90	-10	< 10%
802.11ax@HE_SU 20M_MCS9	-68	-10	< 10%

4.2 Bluetooth RF Specification

Conditions: VDD3.3=3.3V; Ta:25℃	
Features	**Description**
Bluetooth Specification	Bluetooth Low Energy 4.0/4.1/4.2/5.0/5.1/5.2
Frequency Range	2.4~2.4835GHz (2.4GHz ISM Band)
Channels	Bluetooth Low Energy: Ch0~Ch39 (For 2MHz Channels);
Power Classes	Bluetooth Low Energy: Class1.0;
Data Rate & Modulation	LE_1Mbps: GFSK (Uncoded); LE_2Mbps: GFSK (Uncoded); LE_125Kbps: GFSK (Coded_S=8); LE_500Kbps: GFSK (Coded_S=2);
Bluetooth Transmitter Specifications	

Items	Min (dBm)	Typ (dBm)	Max (dBm)
TX Power			
LE_1M/2M	18	20	22
LE_125/500K	18	20	22
Items	Min	Typ	Max
LE_1M Modulation Characteristics			
Δf1avg	225KHz	250.5KHz	275KHz
Δf2avg	185KHz	224.1KHz	275KHz
Δf2max	185KHz	232.8KHz	/
Δf2avg/Δf1avg	0.8	0.89	/
Items	Min	Typ	Max
LE_2M Modulation Characteristics			
Δf1avg	450KHz	500.5KHz	550KHz
Δf2avg	370KHz	436.5KHz	550KHz
Δf2max	370KHz	451.6KHz	/
Δf2avg/Δf1avg	0.8	0.87	/

Bluetooth Receiver Specifications				
Items	Sensitivity		Maximum Input Level	
	Input Level (Typ. dBm)	PER	Input Level (Typ. dBm)	PER
LE_1M	-90	≦ 5%	-10	≦ 5%
LE_2M	-89	≦ 5%	-10	≦ 5%
LE_500K	-94	≦ 5%	-10	≦ 5%
LE_125K	-96	≦ 5%	-10	≦ 5%

4.3 NearLink RF Specification

Conditions: VDD3.3=3.3V; Ta:25℃	
Features	Description
NearLink Technical Requirements	SLE V1.0
Frequency Range	2.4~2.4835GHz (2.4GHz ISM Band)

Channels	Ch0~Ch78 (For SLE_1MHz Channels) Ch2~Ch76 (For SLE_2MHz Channels)
Data Rate & Modulation	SLE_1MHz GFSK; SLE_2MHz GFSK; SLE_1MHz QPSK MCS6; SLE_2MHz QPSK MCS6;

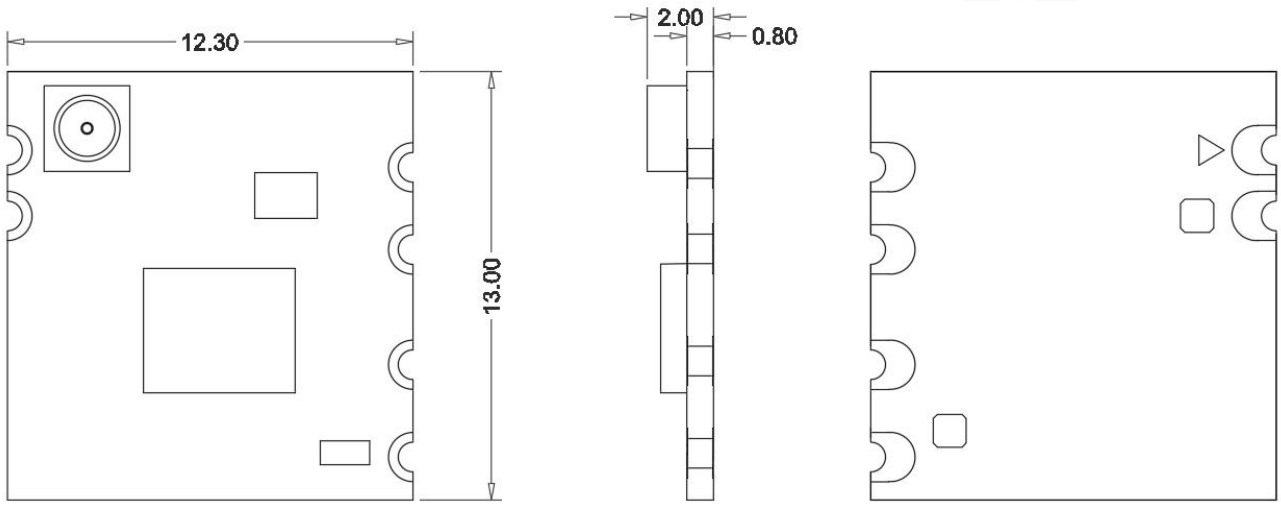
SLE Transmitter Specifications			
Items	**Min（dBm）**	**Typ（dBm）**	**Max（dBm）**
TX Power			
SLE_1M/2MHz GFSK	18	20	22
SLE_1M/2MHz QPSK	12	14	16
Items	**Min**	**Typ**	**Max**
Carrier Freq offset&drift（SLE_1MHz GFSK）			
Freq offset	-150KHz	-1.93KHz	150KHz
Initial freq drift	-50KHz	0.45KHz	50KHz
Items	**Min**	**Typ**	**Max**
Modulation Characteristics（SLE_2MHz GFSK）			
Δf1avg	450KHz	249.6KHz	550KHz
Δf2avg	370KHz	217.6KHz	/
Δf2avg/Δf1avg	0.8	0.87	/
Items	**Min**	**Typ**	**Max**
TX EVM（SLE_2MHz QPSK）			
RMS EVM	/	2.26%	13%
99% EVM	/	4.34%	28%
Peak EVM	/	5.12%	32%
Items	**Min**	**Typ**	**Max**

SLE Receiver Specifications				
Items	**Sensitivity**		**Maximum Input Level**	
	Input Level (Typ. dBm)	**PER**	**Input Level (Typ. dBm)**	**PER**

SLE_1MHz GFSK	-96	≦ 10%	-10	≦ 10%
SLE_2MHz GFSK	-93	≦ 10%	-10	≦ 10%
SLE_1MHz QPSK MCS6	-98	≦ 10%	-10	≦ 10%
SLE_2MHz QPSK MCS6	-95	≦ 10%	-10	≦ 10%

5. Mechanical Specifications

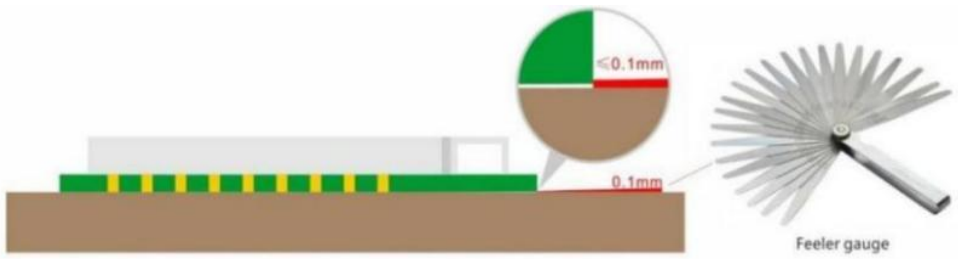
5.1 Module Outline Drawing



(Top View) (Side View) (Bottom View)

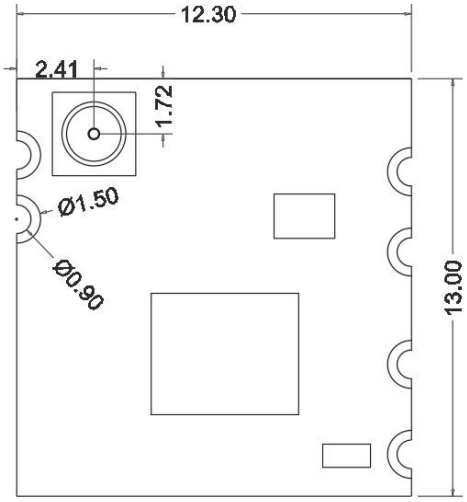
Module dimension: 13.0*12.3*2.0mm (L*W*H; Tolerance: ±0.3mm_L/W, ±0.2mm_H)

IPEX / MHF-1 connector dimension: 3.0*2.6*1.2mm (L*W*H, Ø2.0mm)

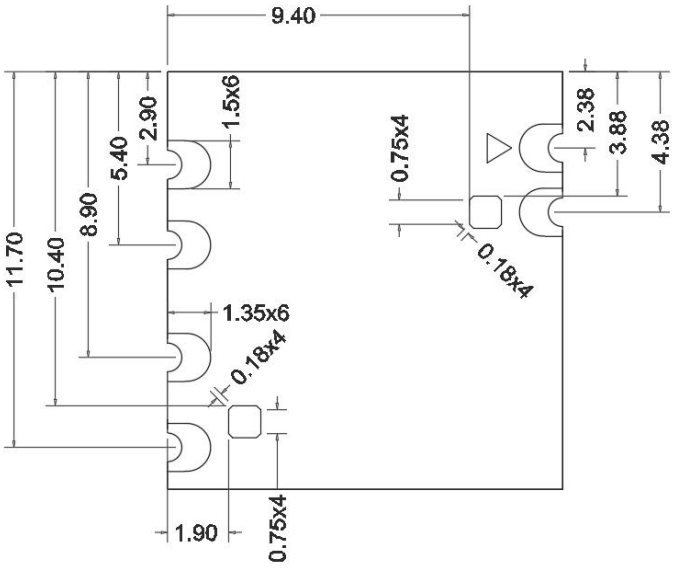


Module Bow and Twist：≤0.1mm

5.2 Mechanical Dimensions



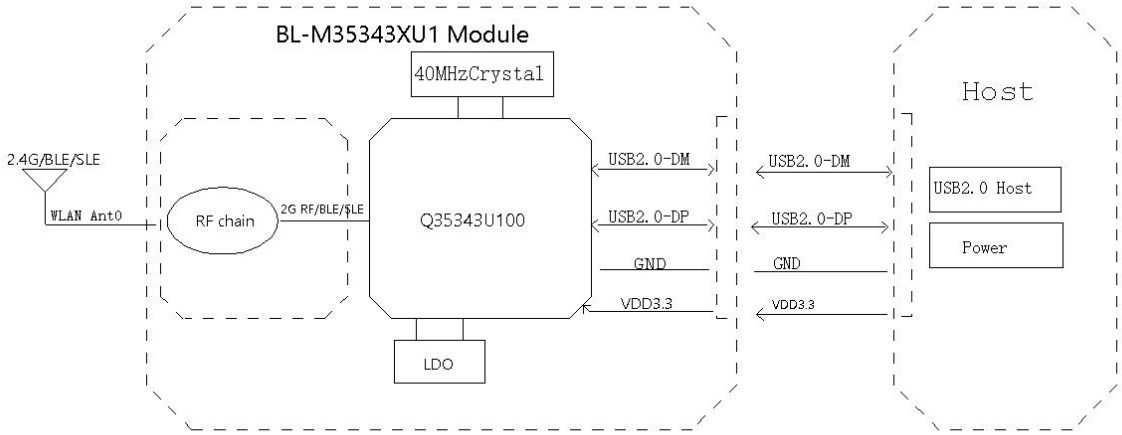
(Top View)



(Bottom View)

6. Application Information

6.1 Typical Application Circuit



6.2 HW Application Note

6.2.1 VDD3.3 Power requirement:

A、DC 3.1~3.5V & Ripple Voltage <100mV power supply input, Maximum RMS current≥500mA and Maximum Peak current≥600mA.

B、For achieve fast transient response，a current mode buck converter recommended.

C、On customer's motherboard, use 10uF and 100nF MLCC capacitors close to the module's VDD3.3 Pin for power input decoupling. Each GND Pin has three close vias to ensure connectivity and thermal conductivity.

6.2.2 USB interface Design Guidelines:

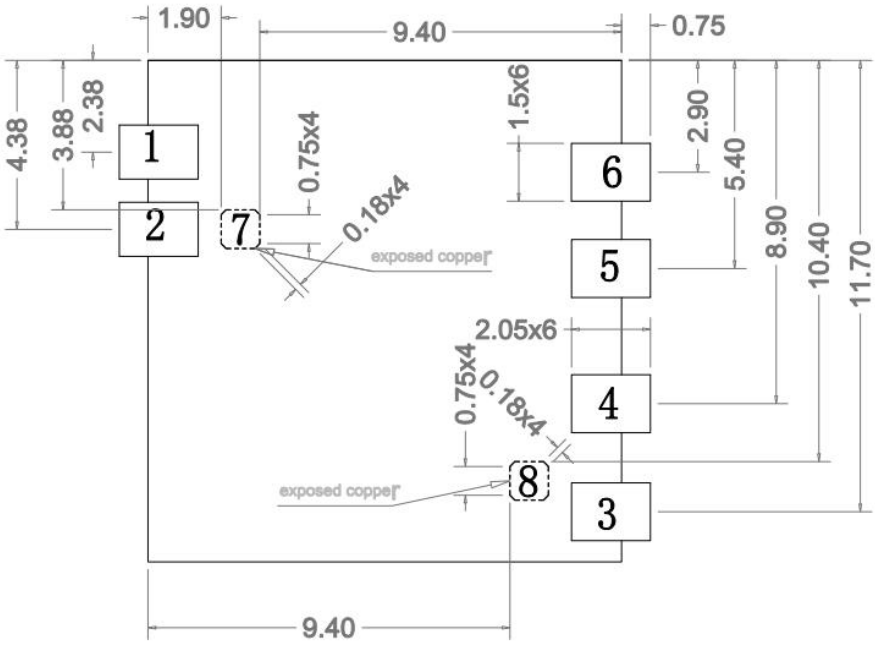
A、The module provides a USB device interface which is compatible with USB2.0 specification, High-Speed mode supports data transmission rate up to 480Mbps.

B、On customer's motherboard, PCB traces of the USB high-speed signal pair should be maintain 90Ω differential impedance, structure of "Differential Coated Coplanar Waveguide With Ground" with the advantages of impedance control and GND surrounding isolation interference may be an ideal choice. To avoid interference, USB signal pair must be far away from power, RF and other signals, GND copper can be used to surround and isolate them.

C、PCB traces of the USB high-speed signal pair as short as possible, as far away from other signal as possible, minimize the length mismatch of signal pair, avoid layer change and maintain a complete reference layer to reduces signal reflections and impedance changes.

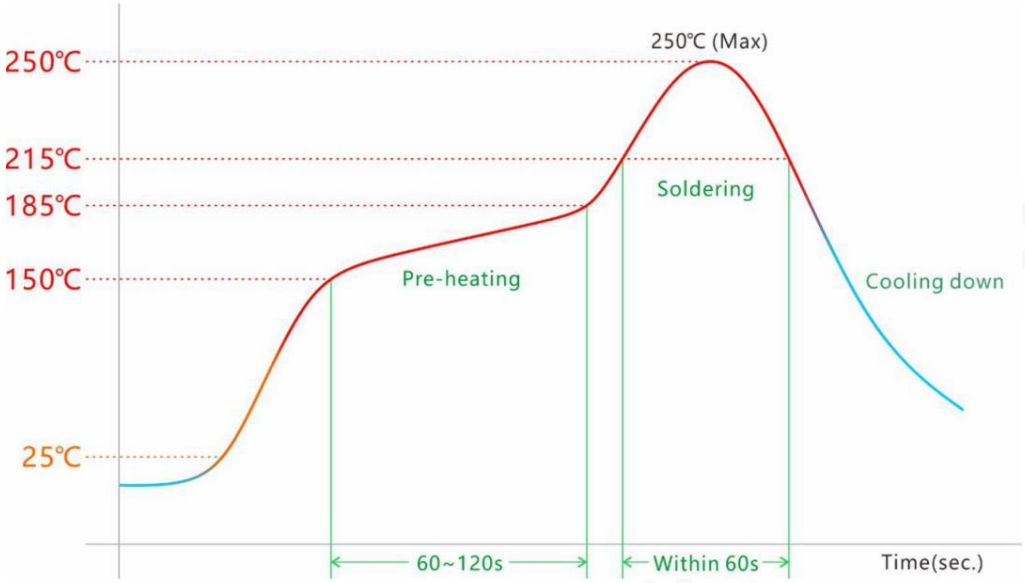
D、If It is necessary to change layers for USB signal pair routing, use GND vias close to signal pair's vias as the shortest return path.

6.3 Recommend PCB Layout Footprint



(Design Units: mm)

6.4 Reflow Soldering Standard Conditions



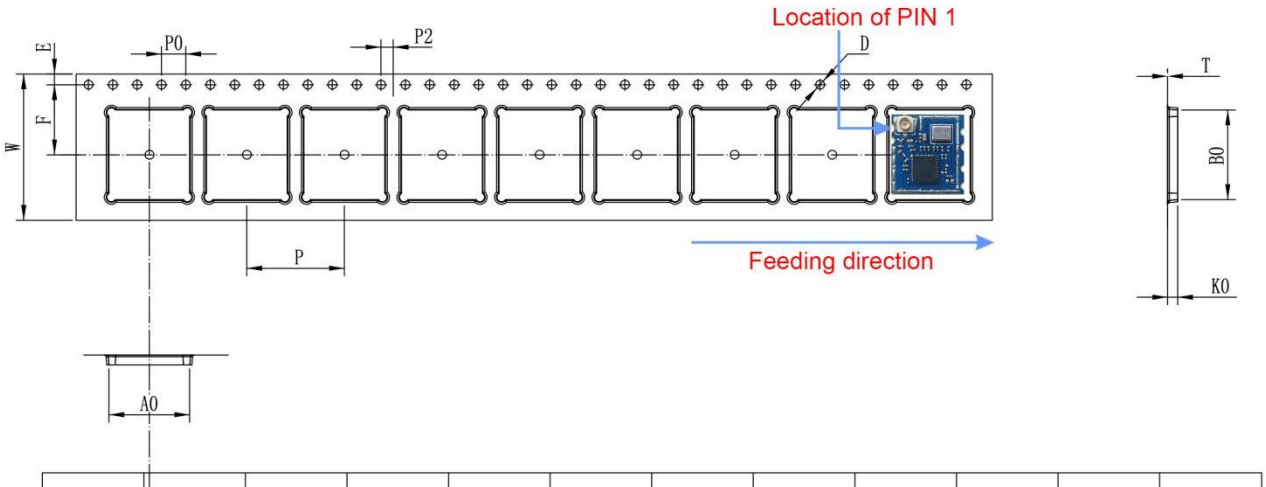
Please use the reflow within 2 times.
Set up the highest temperature within 250℃.

7. Key Components Of Module

No.	Parts	Specification	Manufacturer	Note
1	Chipset	Q35343U100	TriductorTechnology(Suzhou),Inc.	
2	PCB	BL-M35343XU1	SHEN ZHEN QILI ELECTRON CO.,LTD	
			Quzhou Sunlord Electronics Co.,Ltd	
			ShenZhen Tie Fa Technology Limited	
3	Crystal	40MHz-3225	Chengde oscillator Electronic Technology CO.,LTD	
			LUCKI CM ELECTRONICS CO.,LTD	
			JinHua East Crystal Electronic CO.,LTD	

8. Package and Storage Information

8.1 Package Dimensions



ITEM	W	A0	B0	K0	E	F	P	P0	P2	D	T
DIM	24.00+0.3	12.70±0.1	13.60±0.1	2.50±0.1	1.75±0.1	11.5±0.1	20.00±0.1	4.00±0.1	2.00±0.1	Ø1.5±0.1	0.30±0.05



Package specification:

1. 1,000 modules per roll and 5,000 modules per box.
2. Outer box size: 37.5*36*29cm.

3. The diameter of the blue environment-friendly rubber plate is 13 inches, with a total thickness of 28mm (with a width of 24mm carrying belt).
4. Put 1 package of dry agent (20g) and 1 humidity card in each anti-static vacuum bag.
5. Each carton is packed with 5 boxes.

8.2 Storage Conditions

Absolute Maximum Ratings:

Storage temperature: -40℃ to +85℃,
Storage humidity: 10% to 95 (Non-Condensing)

Recommended Storage Conditions:

Storage temperature: 5℃ to +40℃,
Storage humidity: 20% to 90% RH

Please use this Module within 12month after vacuum-packaged.
The Module shall be stored without opening the packing.
After the packing opened, the Module shall be used within 72hours.
When the color of the humidity indicator in the packing changed,
the Module shall be baked before soldering.
Baking condition: 60℃, 24hours, 1time.

ESD Sensitivity:

ESD Protection: 4KV(HBM, Maximum rating)
The Module is a static-sensitive electronic device.
Do not operate or store near strong electrostatic fields.
Take proper ESD precautions!



ESD CAUTION